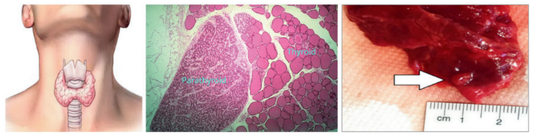
Thyroid Parathyroid (TPT) - Finder

an AI-enabled, real-time surgical tool to eliminate the leading cause of hypoparathyroidism

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Introduction

Over 150,000 thyroidectomies are performed annually in the United States, yet surgeons still rely primarily on visual judgment to identify the tiny and highly variable parathyroid glands. Even for experienced surgeons, these glands can be difficult to distinguish from surrounding tissue, leading to accidental damage or removal in 5–7% of cases. Resulting hypoparathyroidism can cause lifelong calcium imbalance, neurological and cardiac complications, and significant healthcare costs exceeding $280 million annually.



Objective

We are developing a pen-like, AI-enabled multisensing surgical instrument that accurately distinguishes thyroid from parathyroid tissue in real time during surgery. The system includes a multimodal pen-like sensing tip, embedded electronics, on-premises AI inference, and LED-based binary output (thyroid vs parathyroid). Based on our results with ex-vivo chicken muscle tissue, and in-situ and in-vivo pig tissues, we expect our AI-driven, multiple modality approach will yield sensitivity and specificity in the 97-99% range.

Methodology

We are independently engineering and validating three sensing modalities to quantify tissue differences prior to multimodal AI fusion.

1 Bioimpedance

- ✓ Custom multi-frequency impedance measurement PCB
- ✓ Calibrated against structured RC reference networks
- ✓ 500+ ex-vivo tissue measurements collected
- ✓ Magnitude–phase distribution analysis with statistical separation

Quantifiable electrical property differences between tissue types across controlled ionic conditions.

2 Ultrasound

- ✓ 1.8 MHz piezoelectric transducer-based pulse–echo system
- ✓ Dual- and single-transducer architectures evaluated
- ✓ Automated ringdown suppression and waveform segmentation
- ✓ Comparative analysis across media, distance, and pressure conditions

Reproducible, condition-dependent acoustic signatures enabling structural tissue discrimination.

3 Photonics

- ✓ Integrated excitation driver, transimpedance amplifier, band-pass filtering
- ✓ Wavelength-matched LED–photodetector pair for fluorescence capture
- ✓ High-volume acquisition (600+ samples per trial)
- ✓ Signal-to-noise optimization and statistical reliability testing

Robust detection of fluorescence intensity variations relevant to tissue differentiation.

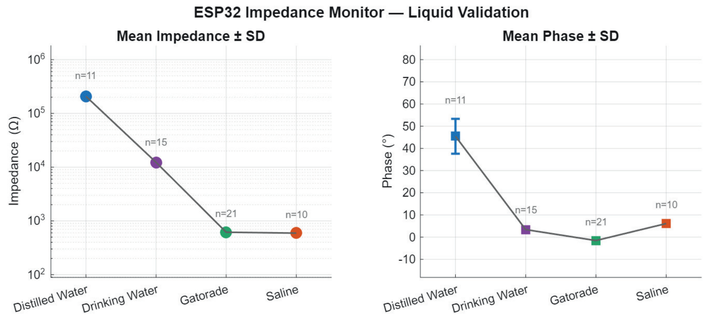


Figure I Mean impedance and phase (33 kHz) for distilled water, drinking water, Gatorade, and saline showing strong conductivity-dependent separation across liquids.

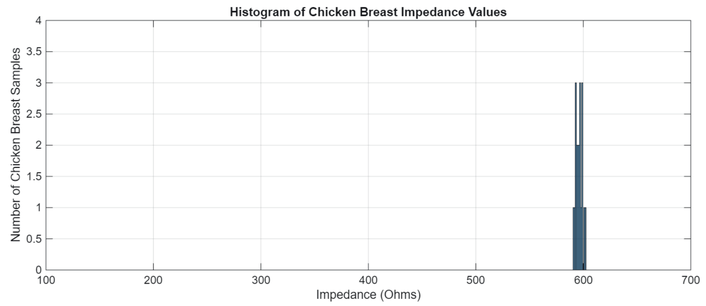


Figure II Histogram of chicken breast impedance measurements (33 kHz) showing tight clustering around ~595 Ω, indicating low measurement variability and stable tissue response.

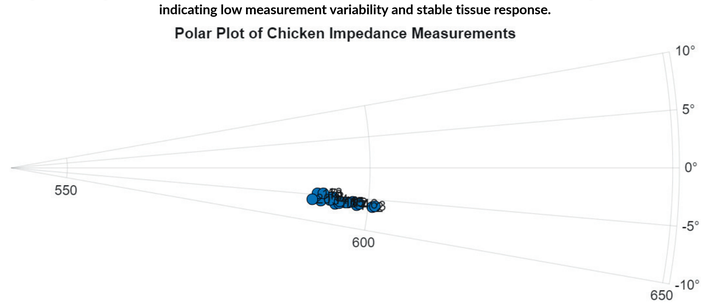


Figure III Polar representation of single-frequency (33 kHz) chicken tissue impedance measurements showing tight clustering, indicating stable and repeatable complex impedance acquisition.

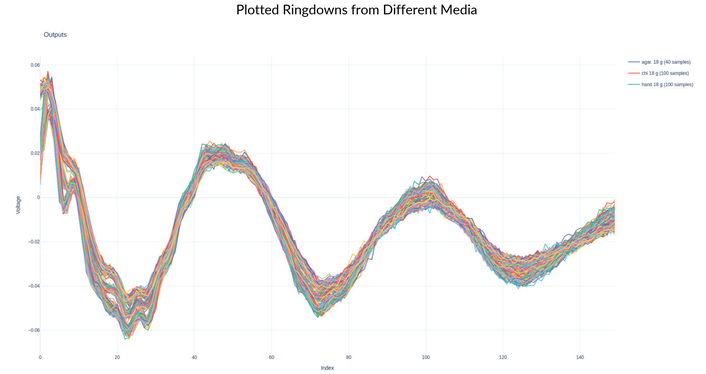


Figure IV Ringdown response of the same transducer touching agar, hand, and chicken thigh under identical pressure. Differences in amplitude and decay rate are caused by each medium's acoustic impedance and internal structure, which affect how ultrasonic energy is absorbed and reflected.

Preliminary Results

Bioimpedance: Electrical features distinguish conditions clearly, exceeding noise levels.

- **2200 Ω calibration achieved 0.034% error at 33 kHz → high complex-impedance accuracy.**
- **Chicken samples showed consistent magnitude with low within-sample variance.**
- **Ionic phantoms (0.1–0.7% NaCl) produced systematic conductivity-dependent shifts.**
- **Tight magnitude–phase clustering confirmed stable, repeatable measurements.**

Ultrasound: Acoustic features encode material-dependent structural differences.

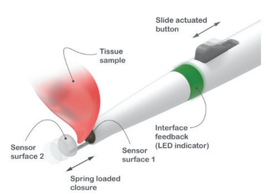
- **Reproducible and distinct waveform differences are observed across different media types**
- **Amplitude and decay trends scale predictably with mechanical loading**

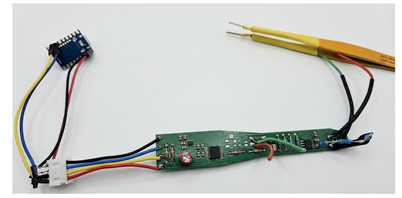
Photonics: Optical intensity provides a robust, statistically separable classification feature.

- **High-volume acquisition (≈600 measurements per trial) demonstrated strong repeatability.**
- **Large effect size (Cohen's d ≈ 1.7), indicating strong signal discrimination relative to noise.**

Multimodal Integration Accuracy

Given the reproducible signal separability observed here and our results with ex-vivo chicken tissues, and in-situ and in-vivo pig tissues, we expect our approach will yield sensitivity and specificity in the 97-99% range following full multimodal AI integration and validation.





Ongoing Work

We have ongoing partnerships with Seattle Children's Hospital and Harborview Medical Center, providing access to animal surgery labs for tissue studies and full system integration.

Bioimpedance

- **Validation: ex-vivo chicken and in-situ/in-vivo pig tissues using tissue mimicking phantoms.**
- **Multi-frequency characterization and supervised ML modeling are being implemented to improve tissue discrimination between fat, muscle, and parathyroid**

Ultrasound

- **Waveform datasets are being shared with the ML team to support model training.**
- **Comparing dual- vs single-transducer architectures for performance vs cost tradeoffs.**

Photonics

- **Optical excitation and detection parameters are being tuned to maximize fluorescence sensitivity and minimize ambient interference.**
- **LED and photodiode data are being collected across tissue types to provide robust training inputs for AI classification.**

Impact & Significance

TPT-Finder represents a new paradigm in surgical decision support: real-time, AI-enabled, multimodal tissue classification at the point of care. Potential impacts includes:

- **Reduction in hypoparathyroidism rates**
- **Decreased hospitalization costs (~$280M annual burden)**
- **Improved surgical safety and outcomes**

With your investment, our work will be able to bridge biomedical sensing, artificial intelligence, human-centered design, and surgical innovation.